H

RAFFLES INSTITUTION

2018 Year 6 Preliminary Examination
Higher 2

BIOLOGY

9744/01

Paper 1 Multiple Choice

25th September 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name and shade your Index Number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions in this paper. Answer all questions. For each question there are four possible answers **A, B, C**, and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

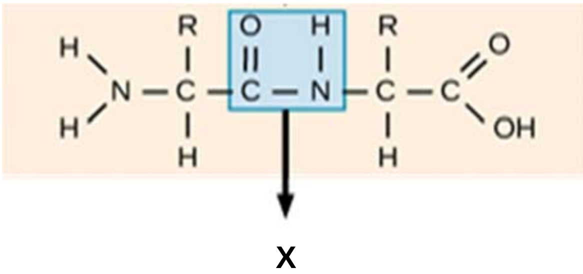
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

(Erase all mistakes completely. Do not bend or fold the OMR Answer Sheet).

This document consists of **21** printed pages.

Raffles Institution
Internal Examination

1. The diagram shows a molecular structure.



What is enclosed by the box X?

A Phosphodiester bond

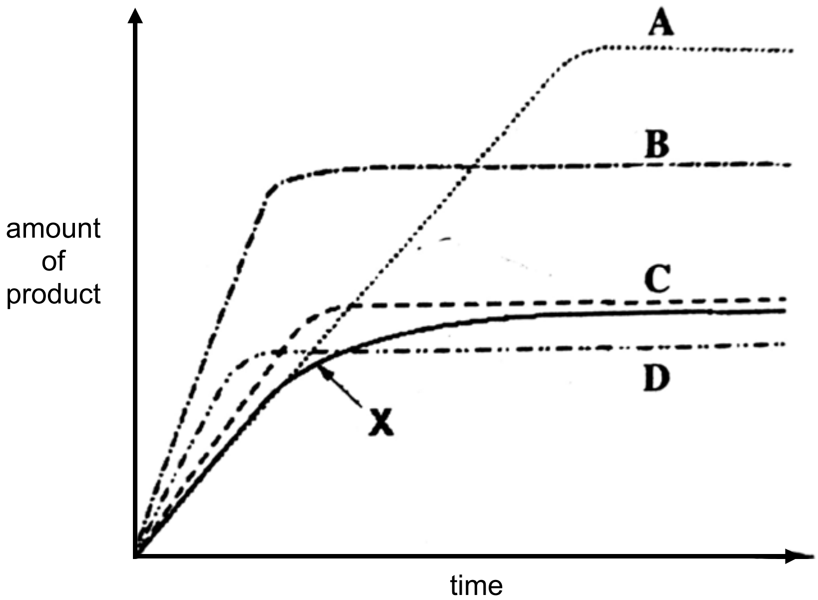
B Glycosidic bond

C Ester bond

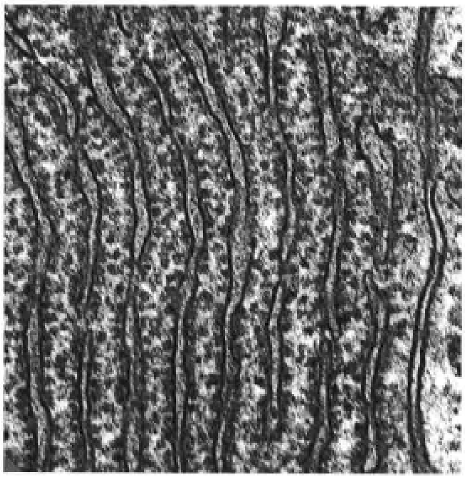
D Peptide bond

2. Line X shows the activity of an enzyme at 20°C. Lines **A** to **D** show the effect of different conditions on the activity of the enzyme.

Which line shows the effect of increasing the temperature by 10°C and adding extra substrate?



3. The electron micrograph shows part of an organelle in a cell.



What describes a function of the cisternae in the organelle shown?

A Moving protein to places where they are covered by phospholipid membranes for secretion outside the cell

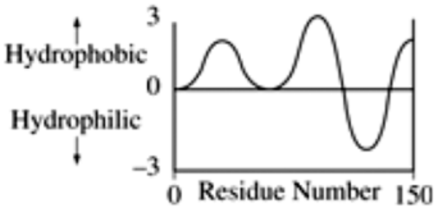
B Producing proteins and covering them with phospholipid membranes for secretion outside the cell

C Producing proteins, covering them with phospholipid membranes and moving them for use inside the cell

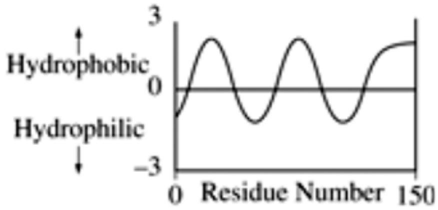
D Producing ribosomes and proteins and storing them in phospholipid membranes for use inside the cell

4. Glycophorin, an integral membrane protein, has a single transmembrane α helix. Which of the following plots most likely represents glycophorin?

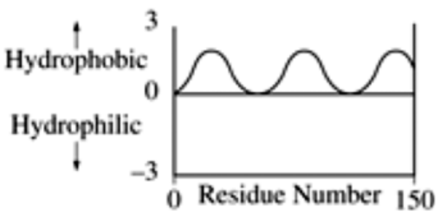
A



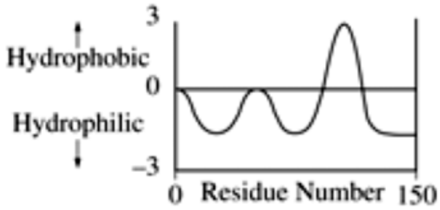
C



B



D



5. A human cell with 44 autosomes, and sex chromosomes X and Y, suffers a non-disjunction at the first meiotic division. Which of the following set of gametes could result?

A

22 + XX | 22 + YY
22 | 22

B

22 + X | 22 + Y
22 + XY | 22

C

22 + XY | 22 + XY
22 | 22

D

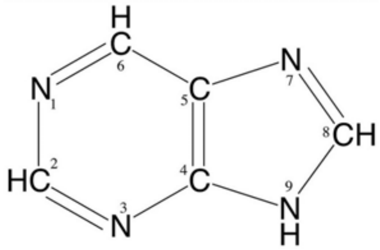
22 + X | 22 + X
22 + Y | 22 + Y

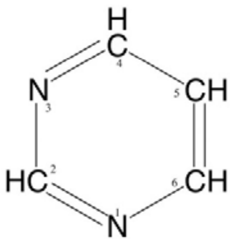
6. The amount of DNA present in the nucleus of a cell at the beginning of interphase is *x* picograms.

Which of the following combinations reflects the amount of DNA, in picograms, in the nucleus during the various stages of cell division?

	End of interphase	End of mitosis	End of meiosis I	Anaphase II
A	x	$\frac{1}{2}x$	$\frac{1}{2}x$	$\frac{1}{2}x$
B	x	$\frac{1}{2}x$	$\frac{1}{2}x$	$\frac{1}{4}x$
C	$2x$	x	x	x
D	$2x$	x	x	$\frac{1}{2}x$

7. The structures of purine and pyrimidine are shown below.

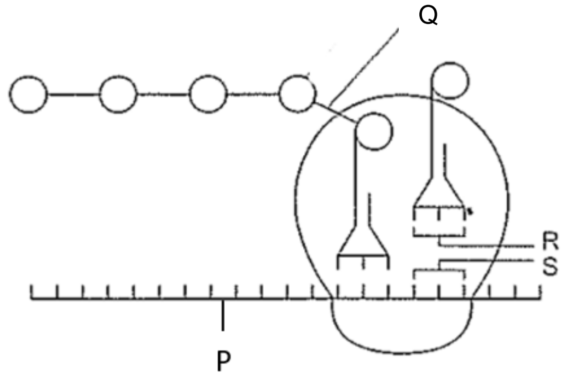




Which of the following correctly shows the number of carbon atoms in the corresponding nucleic acid?

	Molecule	Number of carbons in the molecule
A	DNA strand with the sequence ATCGAAA	33
B	mRNA molecule with the sequence AUCGAAA on 1 strand	30
C	DNA molecule with the sequence ATCGAAA on 1 strand	33
D	DNA strand with the sequence ATCGAAA	68

8. The diagram below shows the process of translation in a prokaryotic cell.



Which of the following correctly identifies the bonds?

	P	Q	Between R and S
A	Peptide	Phosphodiester	Hydrogen
B	Hydrogen	Disulfide	Phosphodiester
C	Phosphodiester	Peptide	Hydrogen
D	Hydrogen	Hydrogen	Peptide

9. The active messenger RNAs (active mRNAs) in tissue cells can be isolated by passing the homogenised cell contents through a fractionating column. The column has short lengths of uracil nucleotides attached to a solid supporting material. Molecules of mRNA that can pass through the column are quickly broken up into small pieces and cannot be translated.

The active mRNAs that attach to the column can be collected subsequently by an appropriate treatment.

Which statements correctly describe active mRNA?

1 Active mRNAs are held to the fractionating column by bonds between adenine and uracil bases.

2 Active mRNAs can be released from the fractionating column by breaking hydrogen bonds.

3 Only mRNAs with polyadenine tailing can be translated.

4 Polyadenine tailing stabilises mRNA and prevents it from being broken up.

A 1 and 2

B 1, 2 and 3

C 3 and 4

D 1, 2, 3 and 4

10. A hybrid phage was artificially created by removing the DNA from lambda phage and replacing it with DNA from T4 phage. This hybrid was allowed to infect a bacterium and reproduce. The progeny of the hybrid phage will have the characteristics of a ____________________.

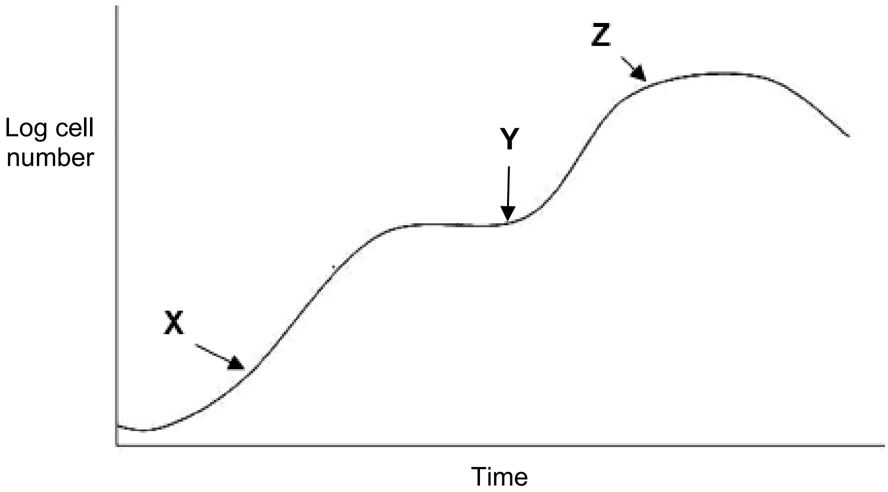
A T4 phage

B lambda phage

C hybrid phage with T4 DNA and lambda proteins

D hybrid virus with lambda DNA and T4 protein

11. *E. coli* bacteria are grown in a culture of nutrients, which includes glucose and lactose as the main source of carbon-based nutrient. The following growth curve is obtained.



Which of the following corresponds correctly to the region specified on the growth curve of *E. coli*?

	CAP activated	High amounts of *lac* polycistronic mRNA	Repressor inactivated
A	X only	X and Y only	Y and Z only
B	Y only	Y and Z only	X, Y and Z
C	Y and Z only	Z only	Y and Z only
D	Y and Z only	Y and Z only	X, Y and Z

12. Which feature occurs in the life cycle of the influenza virus?

A Host cell DNA is destroyed by lytic enzymes.

B Viral genome is integrated into the host genome.

C Viral DNA acts as a template for DNA synthesis.

D Viruses enter the host cell by receptor-mediated endocytosis.

13. Which of the following statements describe possible ways by which viruses can cause disease in animals?

I They inhibit normal host cell DNA, RNA or protein synthesis in host cell.

II They disrupt and inactivate the tumour suppressor genes of the host cell causing uncontrolled cell division.

III They disrupt and inactivate the oncogenes of the host cell causing uncontrolled cell division.

IV Their viral proteins and glycoproteins on the surface membrane of host cells cause them to be recognised and destroyed by the body's immune system.

V They deplete the host cell of cellular materials essential for metabolic functions.

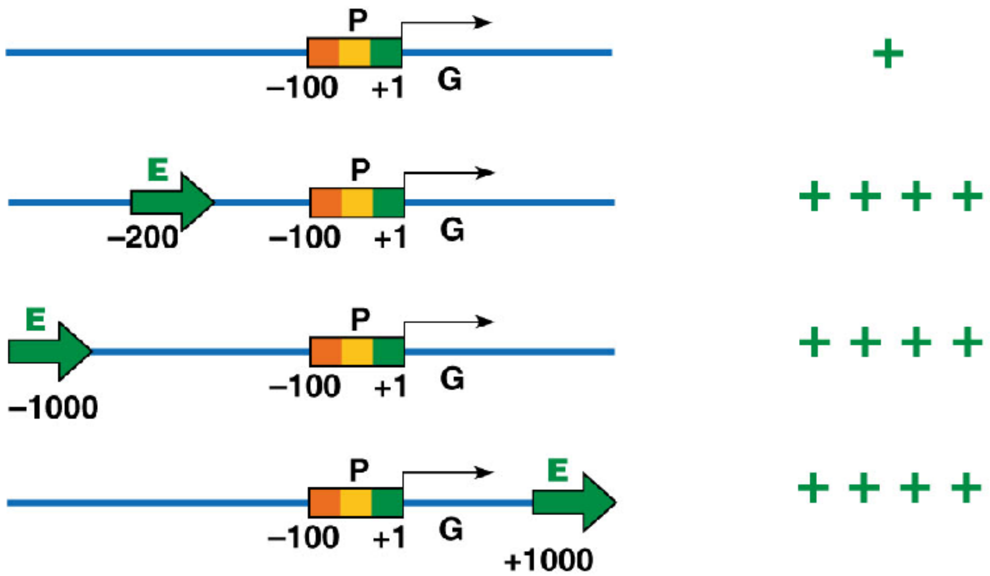
A I, II and V

B I, II and IV

C II, III and V

D I, II, IV and V

14. Recombinant DNA techniques can be used to alter the locations of control elements within DNA sequences, so as to study the effects of these changes on the levels of transcription. The diagram shows the various structures of transcription units and the corresponding relative levels of transcription. The promoter, enhancer and coding sequences are represented by letters P, E and G respectively. The number of symbol '+' indicates the relative frequency of transcription.



With reference to the diagram, which of these statements is a valid conclusion?

A The relative distance between promoter and enhancer has no effect on the frequency of transcription.

B The frequency of transcription is increased when the enhancer is located upstream of the promoter.

C An enhancer is required for transcription.

D Orientation of the enhancer does not affect the frequency of transcription.

15. A geneticist determines that a particular human disease is caused by a gene mutation. The mutant allele contains a substitution of cytosine to adenine at position 334. The DNA sequence for bases 301 to 351 from the non-template strand of the normal allele is shown.

5'- ATG TTA CGA GGT ATC ATA CGA ACG GAG CGC GAA CTA GTT ACT CCC ATA AAA - 3'

Which of the following statements best describes a consequence of this mutation?

A A nonsense mutation has occurred resulting in no protein product being formed.

B The mutant protein contains fewer amino acids than the normal protein.

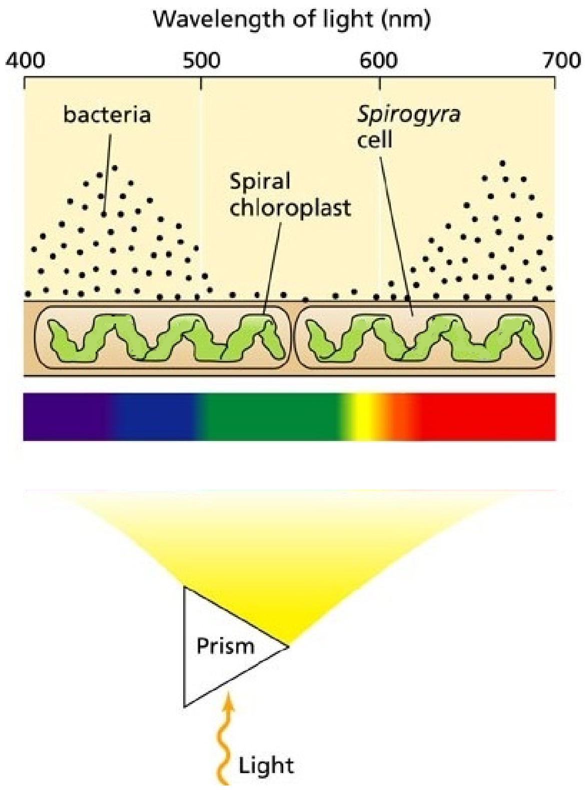
C A missense mutation has occurred resulting in a non-functional protein.

D There is no change in length of amino acid sequence due to the mutation occurring outside of the coding region.

16. In 1882, the German botanist T.W. Engelmann performed an ingenious experiment to investigate the effects of different wavelengths of light on the rate of photosynthesis for a suspension of alga *Spirogyra* (a filamentous microorganism containing long, spiral chloroplasts).

In his experiment, a prism was placed between the light source and the alga filament to produce and scatter all colours of the light across the alga filament evenly. Then, aerobic bacteria were added to the alga filament suspension. All the other variables were kept constant.

After exposure to light for a certain period of time, the bacteria were found to move towards and accumulate at specific lengths along the alga filament as shown below.



Which of the following conclusion(s) can be drawn from the above experiment.

I The directional movement of the bacteria is due to oxygen released from *Spirogyra*.

II Green light is least absorbed, whereas red and blue wavelength of light is efficiently used for photosynthesis.

III NADPH is the reducing power that drives the formation of glyceraldehyde-3-phosphate.

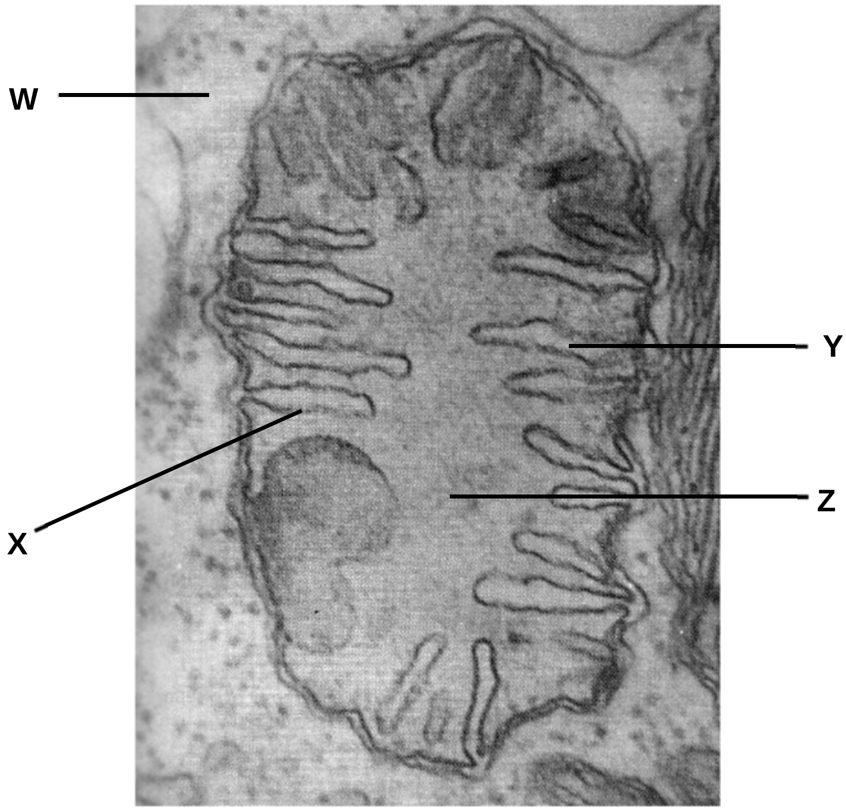
A II only

B I and II only

C II and III only

D All of the above

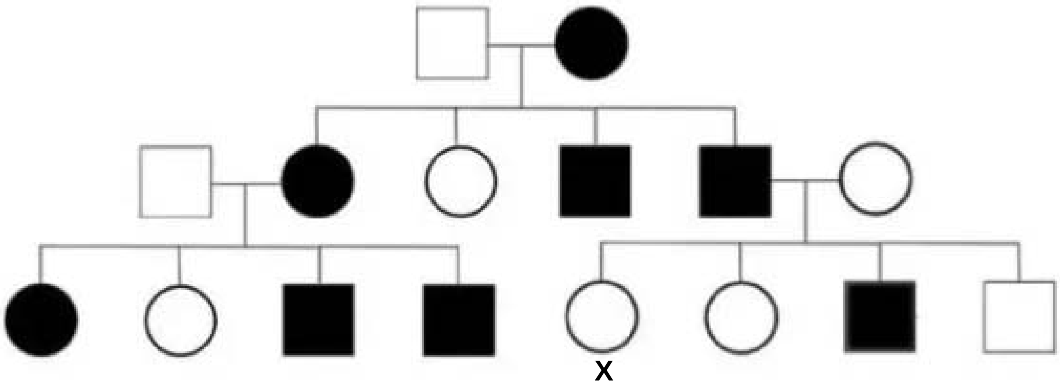
17. The figure below shows an electron micrograph of an organelle.



Which of the following correctly matches the processes with the corresponding structures?

	Formation of pyruvate	Oxidative phosphorylation	Direction of diffusion of H^+ ions	Formation of reduced co-enzymes
A	W	X	Z → Y	Z
B	Z	Y	Z → Y	W and Z
C	Z	Y	Y → Z	W
D	W	X	Y → Z	W and Z

18. The pedigree below shows the inheritance of Marfan syndrome which affects connective tissues in the body.



Individual X is homozygous at the loci for the disease gene. What is the genetic basis of inheritance of the disease?

A Autosomal dominant

B Autosomal recessive

C Sex-linked dominant

D Sex-linked recessive

19. Coat colour in mice is controlled by two genes, each with two alleles. The genes are on different chromosomes.

One gene controls pigment colour. The presence of allele **A** results in a yellow and black banding pattern on individual hairs, producing an overall grey appearance called agouti. Mice with the genotype **aa** do not make the yellow pigment and are, therefore, black.

The other gene determines whether any pigment is produced. The allele **D** is required for development of coat colour. Mice with the genotype **dd** produce no pigment and are called albino.

An albino mouse is mated with a black mouse to produce 12 albino mice, 7 agouti mice and 5 black mice.

A χ^2 test was performed to test the significance of the difference between the observed and expected results.

distribution of χ^2			
number of degrees of freedom (v)	probability		
	0.1	0.05	0.01
1	2.71	3.84	6.64
2	4.60	5.99	9.21
3	6.25	7.82	11.34
4	7.78	9.49	13.28

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

O = observed result
E = expected result
v = n–1

Using the equation and the table of χ^2 values, which of the following correctly describes the result of the χ^2 test?

	number of degrees of freedom (v)	probability	significance of difference between observed and expected results
A	2	< 0.05	significant
B	3	< 0.05	significant
C	2	> 0.05	not significant
D	3	> 0.05	not significant

20. Pure breeding plants of contrasting traits were cross fertilised and the seeds were planted in pots of soil containing equal proportion of fertiliser. The pots were then exposed to different light conditions for 60 days. Throughout the investigation, the plants were watered with equal amount of water, twice daily.

At the end of the investigation, the plants' height, number of leaves, length of leaves and colour of leaves were measured and summarised in the table below.

	No light	Dim Light	Bright light
Height/cm	10.3 ± 0.3	8.1 ± 0.5	6.6 ± 0.4
Length of leaves/cm	1.7 ± 0.3	1.7 ± 0.2	1.6 ± 0.1
Colour of leaves	Yellow	Pale green	Dark green

Which of the following statement(s) cannot be explained by the data?

1 The additive effect of genes is responsible for the continuous variation observed in the height and length of leaves.

2 The genes involved in chlorophyll pigment synthesis are activated by light.

3 Leaf colour is controlled by a gene locus whereby heterozygotes have pale green leaves.

A 2 only

B 3 only

C 1 and 3 only

D 2 and 3 only

21. Totipotency is demonstrated when __________.

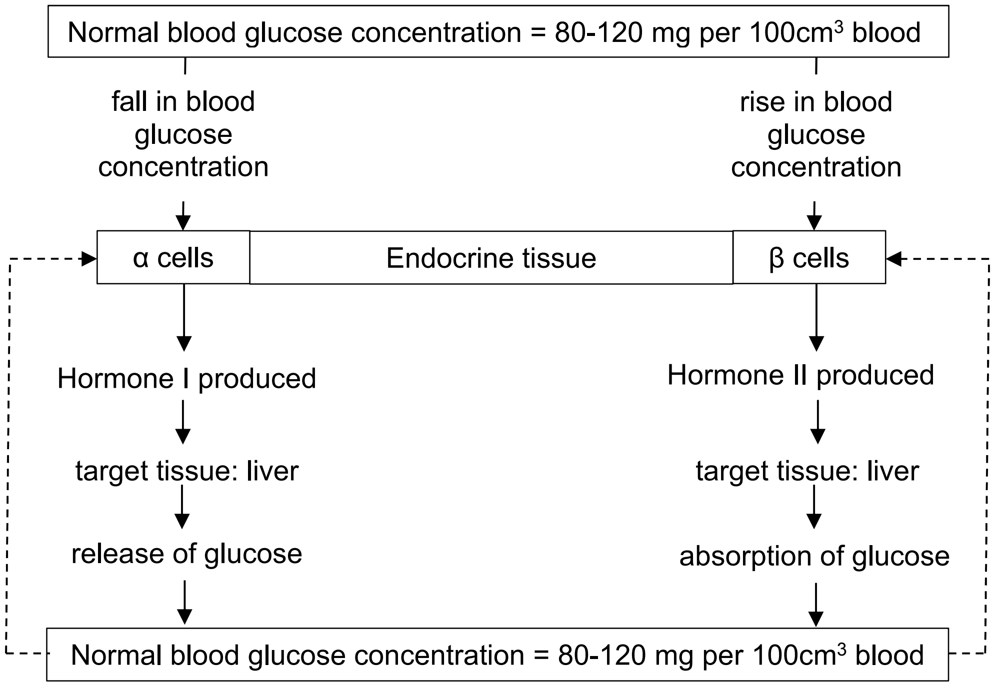
A cancer cells give rise to heterogeneous cell types

B a stem cell can differentiate into placental cells and all cells in an organism

C a hematopoietic stem cell differentiates into a lymphocyte

D an embryonic stem cell divides and differentiates

22. The diagram below shows the role of an endocrine tissue in controlling blood glucose concentration.



Which of the following statements are true?

1 Small amounts of hormone I and II inducing a large response in liver tissue demonstrates positive feedback.

2 Hormones I and II inducing different responses from the same target tissue is due to the hormones binding to different receptors on the liver cell surface membrane.

3 The binding of hormone I to receptors on liver cell surface membrane leads to the production of second messenger, cAMP.

4 Besides the liver, hormone I will also target muscle tissue to regulate blood glucose concentration.

A 1 and 2

B 2 and 3

C 3 and 4

D 2 and 4

23. Which of the following statements is false about cell signalling involving tyrosine kinase receptors?

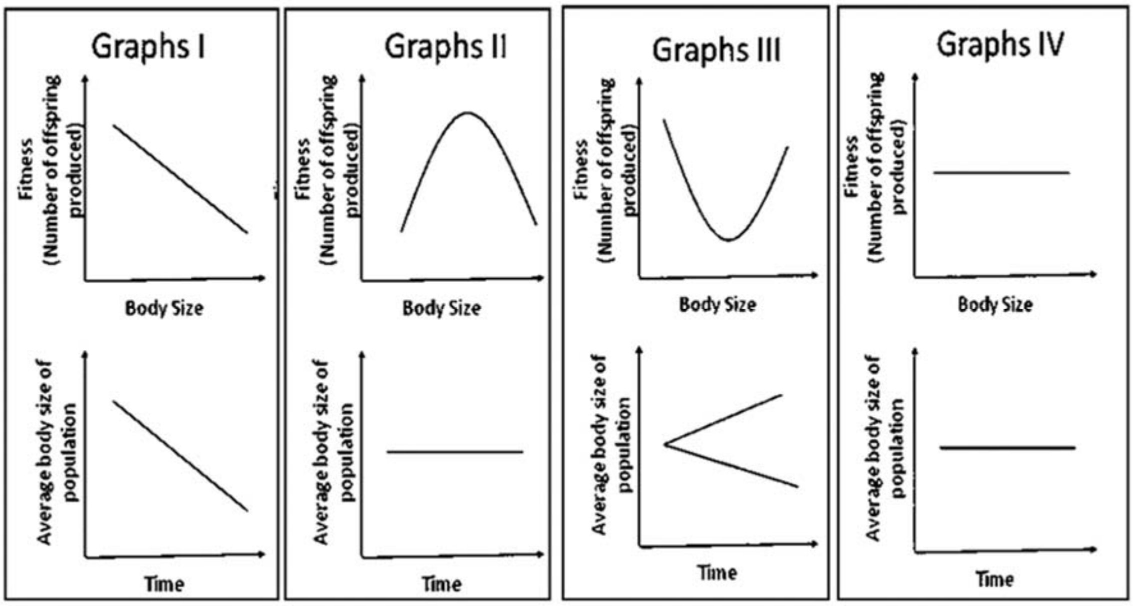
A Ligand molecules are mostly hydrophilic in nature.

B Different activated relay proteins serve to directly amplify the effects of the ligand.

C Dimerisation serves to initiate auto-phosphorylation.

D Receptors are transmembrane proteins that are anchored within the cell surface membrane.

24. The different forms of natural selection can be distinguished according to their effect on the body size of the pink salmon (*Onchorhynchus gorbuscha*).



Which of the following describes the correct form of natural selection for each of the following sets of graphs?

	Graphs I	Graphs II	Graphs III	Graphs IV
A	Disruptive selection	Directional selection	Stabilising selection	No selection
B	No selection	Stabilising selection	Directional selection	Disruptive selection
C	Directional selection	Stabilising selection	Disruptive selection	No selection
D	Directional selection	Disruptive selection	No selection	Stabilising selection

25. Bacteria in the genus *Wolbachia* infect many butterfly species. They are passed from one generation to the next in eggs, but not in sperms, and they selectively kill developing male embryos.

During the 1960s in Samoa, the proportion of male blue moon butterflies fell to less than 1% of the population. However, by 2006, the proportion of males was almost 50% of the population.

Resistance to *Wolbachia* is the result of the dominant allele of a suppressor gene.

Which statements correctly describe the evolution of resistance to *Wolbachia* in the blue moon butterfly population?

I *Wolbachia* acts as a selective agent.

II The selective killing of male embryos is an example of artificial selection.

III When infected with *Wolbachia,* male embryos that are homozygous for the recessive allele of the suppressor gene die.

IV All male embryos that carry the dominant allele of the suppressor gene pass that allele to their offspring.

V The frequency of the dominant allele of the suppressor gene rises in the butterfly population.

A I and IV

B II and III

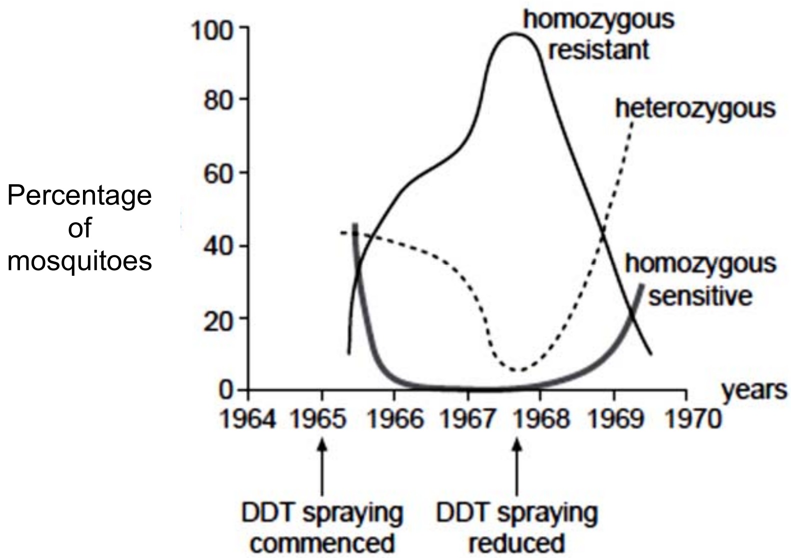
C I, III and V

D II, IV and V

26. In the mid-1960s, DDT was widely used as an insecticide against mosquitoes. The sensitivity to insecticide in mosquitoes is determined by a single gene that has two alleles.

allele 1 : resistant to DDT
allele 2 : sensitive to DDT

Over several years, genotypic frequencies were measured in a population of mosquito larvae. The graph below shows the results.



Analysis of the graph reveals that in the population, ___________________________ .

A allele 1 confers a selective disadvantage in the absence of DDT.

B heterozygote advantage is demonstrated after DDT spraying is reduced.

C mutant allele 1 emerged as a result of the use of DDT in 1965.

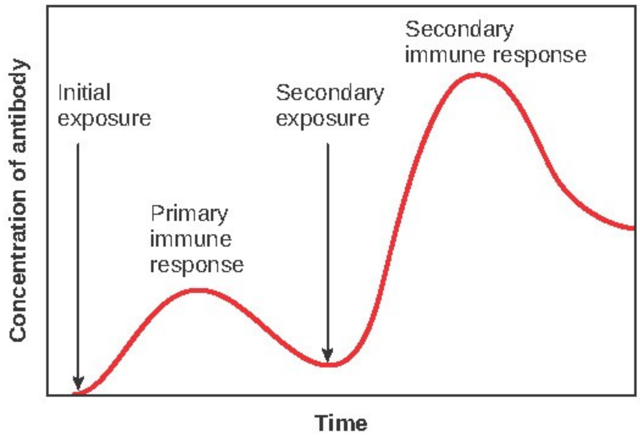
D only one copy of allele 1 is required for resistant phenotype.

27. The diagram below shows the structure of an antibody.

Which of the following correctly matches the events with the regions in which diversity is generated?

	Somatic recombination	Somatic hypermutation	Class switching
A	X	X	Z
B	Y	X	Z
C	Y	Y	Z
D	Z	X	Y

28. The diagram below shows the antibody production during a primary and a secondary immune response.



Which of the following statement(s) is/are correct?

I Class switching only occurs during the secondary immune response.

II The secondary immune response is faster and stronger compared to the primary immune response.

III Class switching results in the production of antibodies with higher binding affinity during the secondary immune response.

IV Vaccination 'primes' the immune system such that a secondary immune response can be mounted when the body encounters the actual pathogen.

A II only

B II and IV only

C I, II and IV only

D All of the above

29. Which of the following describes a positive feedback concerning climate change?

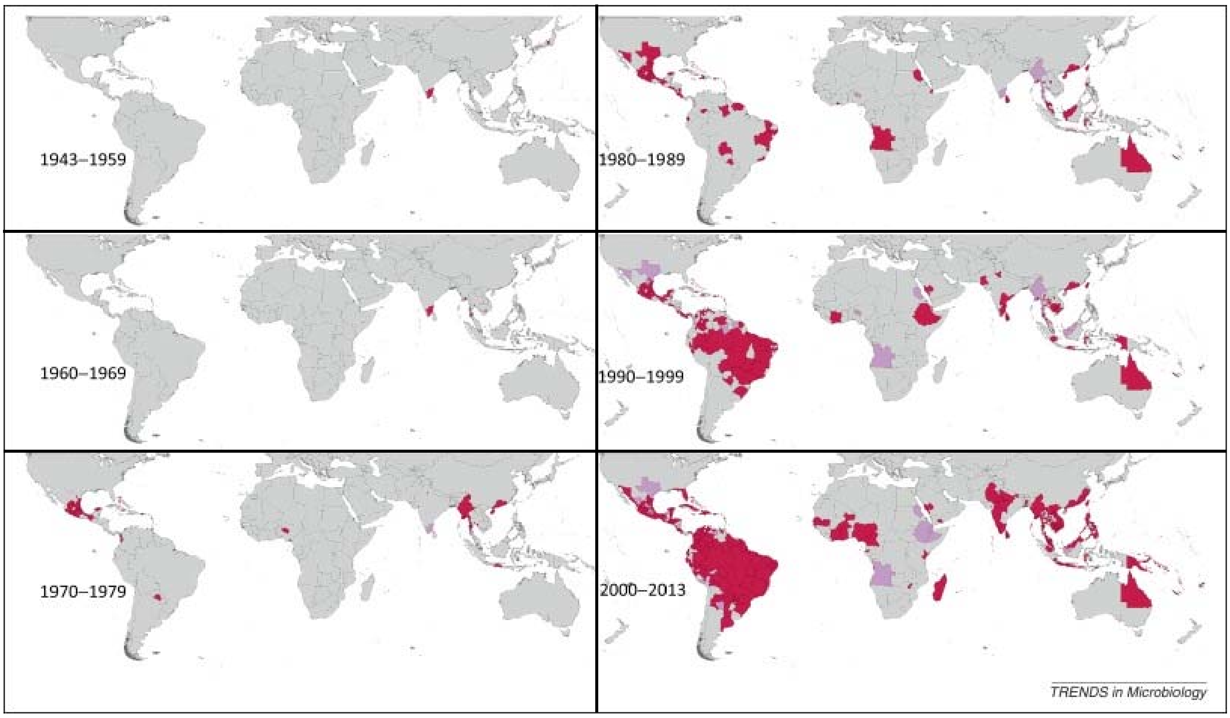
A Increased atmospheric temperature result in melting of sea ice which decreases the amount of sunlight reflected back into space.

B Increased burning of fossil fuels increases atmospheric CO_2 concentration, enhancing the greenhouse effect.

C Melting of glaciers causes an increase in sea levels.

D Increase in atmospheric temperature causes many species to move towards increased altitudes to stay within their optimum temperature range.

30. The diagram below shows the distribution of confirmed cases of dengue fever from 1943 to 2013.



Which of the following explain the observed changes in distribution?

- I Increased global human traffic.
- II Increased global temperatures allow mosquitoes to survive better at increased latitudes.
- III Increased global temperatures allow mosquitoes carrying the dengue virus to move northwards.
- IV Increased global temperatures increases the replication rate of the dengue virus in mosquitoes.
- V Increased global temperatures reduces the replication rate of the dengue virus in humans.

A II and V only

B II, III and IV only

C I, II and IV only

D All of the above

- End of Paper -

Answers:

1.	**D**	**7.**	**D**	**13.**	**D**	**19.**	**C**	**25.**	**C**
2.	**B**	**8.**	**C**	**14.**	**A**	**20.**	**C**	**26.**	**A**
3.	**A**	**9.**	**D**	**15.**	**C**	**21.**	**B**	**27.**	**A**
4.	**D**	**10.**	**A**	**16.**	**B**	**22.**	**B**	**28.**	**B**
5.	**C**	**11.**	**D**	**17.**	**D**	**23.**	**B**	**29.**	**A**
6.	**C**	**12.**	**D**	**18.**	**A**	**24.**	**C**	**30.**	**C**